Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Nutcher Farms, LLC

Physical address of dairy:

5213 West Grayson RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X050-X004-XXXX X017-X050-X005-XXXX

B. OPERATORS

Nutcher, Rick

Operator name: Nutcher, Rick

Telephone no.: Landline: ; Cellular: (209) 652-5185

P.O. Box 268	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Nutcher, Rick

Legal owner name: Nutcher, Rick

Telephone no.: Landline: ; Cellular: (209) 652-5185

P.O. Box 268	Denair	CA	95316
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	360	253	0	0
Number under roof	700	59	0	0	0	0
Maximum number	700	59	360	253	0	0
Average number	700	59	360	253	0	0
Avg live weight (lbs)	1,350	1,200	1,050	700		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,803.22 *tons per reporting period*

Total nitrogen from manure: 314,448.49 *lbs per reporting period*

After ammonia losses (30% loss applied): 220,113.94 *lbs per reporting period*

Total phosphorus from manure: 52,985.59 *lbs per reporting period*

Total potassium from manure: 138,655.46 *lbs per reporting period*

Total salt from manure: 343,162.05 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 29,415,000 *gallons*

Total nitrogen generated: 93,810.20 *lbs*

Total phosphorus generated: 21,050.60 *lbs*

Total potassium generated: 257,080.41 *lbs*

Total salt generated: 1,186,165.56 *lbs*

	29,415,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	29,415,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW	Ground water
Irrigation Water - Canal Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/04/2023	Liquid commercial fertilizer *UN32*	486.82 *gal*			32.000000	0.000000	0.000000	0.000000
07/10/2023	Liquid commercial fertilizer *UN32*	337.03 *gal*			32.000000	0.000000	0.000000	0.000000
08/01/2023	Liquid commercial fertilizer *UN32*	486.82 *gal*			32.000000	0.000000	0.000000	0.000000
08/12/2023	Liquid commercial fertilizer *UN32*	337.03 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	4,400.02	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	4,400.02	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Almonds 113	113	113	0	none	X017-X050-X001-XXXX
Almonds 28	28	28	0	none	X017-X051-X072-XXXX
Almonds 60	60	60	0	none	X017-X050-X006-XXXX
Cherry 35	35	35	0	none	X017-X051-X015-XXXX
Cherry 53	53	53	0	none	X017-X051-X017-XXXX
Field 1	26	26	3	manure	X017-X050-X004-XXXX
Field 2	22	22	2	both	X017-X050-X004-XXXX
Field 3	35	35	3	both	X017-X050-X004-XXXX
Field 4	62	62	3	both	X017-X050-X005-XXXX
Field 5	43	43	3	both	X017-X050-X005-XXXX
Field 6	62	62	3	both	X017-X050-X005-XXXX
Field 7	18	18	3	manure	X017-X050-X005-XXXX
Totals for areas that were used for application	268	268	20		
Totals for areas that were not used for application	289	289	0		
Land application area totals	557	557	20		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 26 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	426.32 *ton*	Dry-weight		68.0	14,800.00	2,820.00	16,800.00		7.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	16.40	155.31	29.59	176.30	786.00

Field 1

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	856.32 *ton*	Dry-weight		70.5	11,600.00	1,850.00	18,500.00		7.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.94	225.41	35.95	359.49	1,455.45

09/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 26 Plant date: 09/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	213.23 *ton*	Dry-weight		66.8	15,300.00	3,010.00	31,800.00		20.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	8.20	83.32	16.39	173.17	1,116.34

Field 2

Field name: Field 2

11/05/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 22 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	219.89 *ton*	Dry-weight		59.2	35,200.00	5,180.00	43,700.00		17.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	10.00	287.09	42.25	356.41	1,451.75

Field 2

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	704.32 *ton*	Dry-weight		69.0	12,200.00	2,200.00	17,700.00		7.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.01	242.16	43.67	351.33	1,498.60

Field 3

Field name: Field 3

11/05/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 35 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	349.85 *ton*	Dry-weight		55.0	31,100.00	5,340.00	46,200.00		17.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	186.00	27.60	252.00	0.00
Total actual harvest content	10.00	279.78	48.04	415.62	1,574.33

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,078.62 *ton*	Dry-weight		67.0	12,400.00	1,850.00	13,900.00		6.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.82	252.21	37.63	282.72	1,320.05

Field 3

09/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 35 Plant date: 09/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	349.56 *ton*	Dry-weight		71.4	14,700.00	1,830.00	11,900.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.99	83.98	10.45	67.98	776.94

Field 4

Field name: Field 4

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 62 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	992.10 *ton*	Dry-weight		67.6	15,800.00	3,000.00	18,300.00		7.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	16.00	163.83	31.11	189.75	774.57

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,612.32 *ton*	Dry-weight		62.2	12,600.00	1,800.00	14,400.00		6.69

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.01	247.71	35.39	283.10	1,315.25

Field 4

09/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 62 Plant date: 09/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	512.32 *ton*	Dry-weight		69.4	16,500.00	2,460.00	9,930.00		18.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	8.26	83.44	12.44	50.22	945.68

Field 5

Field name: Field 5

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 43 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	654.23 *ton*	Dry-weight		67.0	16,300.00	3,050.00	19,000.00		7.88

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	15.21	163.68	30.63	190.79	791.28

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,380.52 *ton*	Dry-weight		71.0	12,500.00	2,310.00	16,500.00		6.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.11	232.76	43.01	307.25	1,169.40

Field 5

09/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 43 Plant date: 09/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	344.10 *ton*	Dry-weight		63.8	16,200.00	3,610.00	31,900.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	8.00	93.86	20.92	184.82	857.47

Field 6

Field name: Field 6

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 62 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	941.23 *ton*	Dry-weight		61.2	15,200.00	2,880.00	16,800.00		7.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	15.18	179.06	33.93	197.91	844.67

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	1,698.52 *ton*	Dry-weight		64.0	12,700.00	1,960.00	17,600.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.40	250.50	38.66	347.16	1,499.08

Field 6

09/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 62 Plant date: 09/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	601.32 *ton*	Dry-weight		68.6	14,100.00	2,920.00	33,900.00		15.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.70	85.88	17.79	206.48	950.16

Field 7

Field name: Field 7

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 18 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/10/2023	287.98 *ton*	Dry-weight		68.4	15,200.00	2,730.00	16,000.00		6.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	16.00	153.69	27.60	161.78	692.62

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	589.52 *ton*	Dry-weight		71.4	12,700.00	2,260.00	17,700.00		7.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.75	237.92	42.34	331.59	1,472.46

Field 7

09/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 18 Plant date: 09/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	129.58 *ton*	Dry-weight		54.2	14,400.00	2,700.00	32,800.00		26.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.20	94.96	17.80	216.29	1,714.49

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 1

Crop: Wheat, silage, soft dough — Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	201.35	65.33	836.50	0.00	410.00 *ton*
Application event totals		201.35	65.33	836.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	6.05	0.00	0.00	0.00	5,152,275.00 *gal*
Application event totals		6.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	5.97	0.00	0.00	0.00	5,084,925.00 *gal*
Application event totals		5.97	0.00	0.00	0.00	

Field 1 - 05/22/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	181.70	58.96	754.89	0.00	370.00 *ton*
Application event totals		181.70	58.96	754.89	0.00	

Field 1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.23	0.00	0.00	0.00	3,603,225.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.15	0.00	0.00	0.00	3,535,875.00 *gal*
Application event totals		4.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Irrigation Water - Canal Water	Surface water	4.23	0.00	0.00	0.00	3,603,225.00 *gal*
Application event totals		54.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.23	0.00	0.00	0.00	3,603,225.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.15	0.00	0.00	0.00	3,535,875.00 *gal*
Application event totals		4.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Irrigation Water - Canal Water	Surface water	4.07	0.00	0.00	0.00	3,468,525.00 *gal*
Application event totals		54.07	0.00	0.00	0.00	

Field 1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.23	0.00	0.00	0.00	3,603,225.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Field 1 - 09/22/2023: Sudangrass, silage

Field name: Field 1

Crop: Sudangrass, silage

Plant date: 09/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	103.13	33.46	428.45	0.00	210.00 *ton*
Application event totals		103.13	33.46	428.45	0.00	

09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.23	0.00	0.00	0.00	3,603,225.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.07	0.00	0.00	0.00	3,468,525.00 *gal*
Application event totals		4.07	0.00	0.00	0.00	

Field 2 - 11/05/2022: Ryegrass, hay

Field name: Field 2

Crop: Ryegrass, hay

Plant date: 11/05/2022

Field 2 - 11/05/2022: Ryegrass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	189.64	22.19	347.08	1,982.50	732,000.00 *gal*
Irrigation Water - Canal Water	Surface water	7.01	0.00	0.00	0.00	5,051,250.00 *gal*
Application event totals		196.66	22.19	347.08	1,982.50	

03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	147.48	20.51	303.15	1,649.58	720,000.00 *gal*
Irrigation Water - Canal Water	Surface water	6.87	0.00	0.00	0.00	4,950,225.00 *gal*
Application event totals		154.35	20.51	303.15	1,649.58	

Field 2 - 05/22/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	203.13	65.91	843.92	0.00	350.00 *ton*
Application event totals		203.13	65.91	843.92	0.00	

05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.04	10.97	126.43	513.22	330,000.00 *gal*
Irrigation Water - Canal Water	Surface water	4.77	0.00	0.00	0.00	3,434,850.00 *gal*
Application event totals		38.80	10.97	126.43	513.22	

Field 2 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.91	0.00	0.00	0.00	3,535,875.00 *gal*
Application event totals		4.91	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.34	11.72	109.11	370.27	306,000.00 *gal*
Irrigation Water - Canal Water	Surface water	4.68	0.00	0.00	0.00	3,367,500.00 *gal*
Application event totals		27.02	11.72	109.11	370.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.72	0.00	0.00	0.00	3,401,175.00 *gal*
Application event totals		4.72	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.94	9.97	114.93	466.56	300,000.00 *gal*
Irrigation Water - Canal Water	Surface water	4.63	0.00	0.00	0.00	3,333,825.00 *gal*
Application event totals		35.57	9.97	114.93	466.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.82	0.00	0.00	0.00	3,468,525.00 *gal*
Application event totals		4.82	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.68	0.00	0.00	0.00	3,367,500.00 *gal*
Application event totals		4.68	0.00	0.00	0.00	

Field 3 - 11/05/2022: Ryegrass, hay

Field name: Field 3

Crop: Ryegrass, hay

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	158.36	22.02	296.20	1,771.33	1,230,000.00 *gal*
Irrigation Water - Canal Water	Surface water	6.67	0.00	0.00	0.00	7,644,225.00 *gal*
Application event totals		165.03	22.02	296.20	1,771.33	

03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	181.74	21.26	332.61	1,899.86	1,116,000.00 *gal*
Irrigation Water - Canal Water	Surface water	6.61	0.00	0.00	0.00	7,576,875.00 *gal*
Application event totals		188.35	21.26	332.61	1,899.86	

Field 3 - 05/22/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	138.63	44.98	575.93	0.00	380.00 *ton*
Application event totals		138.63	44.98	575.93	0.00	

05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.17	9.40	108.37	439.90	450,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.38	0.00	0.00	0.00	6,162,525.00 *gal*
Application event totals		34.55	9.40	108.37	439.90	

Field 3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.32	0.00	0.00	0.00	4,950,225.00 *gal*
Application event totals		4.32	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.00	0.00	0.00	0.00	4,579,800.00 *gal*
Application event totals		4.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.17	7.95	117.51	639.41	444,000.00 *gal*
Irrigation Water - Canal Water	Surface water	3.79	0.00	0.00	0.00	4,344,075.00 *gal*
Application event totals		60.96	7.95	117.51	639.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.06	0.00	0.00	0.00	4,647,150.00 *gal*
Application event totals		4.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.48	8.27	122.27	665.33	462,000.00 *gal*
Irrigation Water - Canal Water	Surface water	4.41	0.00	0.00	0.00	5,051,250.00 *gal*
Application event totals		63.89	8.27	122.27	665.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.38	0.00	0.00	0.00	5,017,575.00 *gal*
Application event totals		4.38	0.00	0.00	0.00	

Field 3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.00	0.00	0.00	0.00	4,579,800.00 *gal*
Application event totals		4.00	0.00	0.00	0.00	

Field 3 - 09/22/2023: Sudangrass, silage

Field name: Field 3

Crop: Sudangrass, silage

Plant date: 09/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.22	26.87	250.12	848.82	1,116,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.55	0.00	0.00	0.00	6,364,575.00 *gal*
Application event totals		56.78	26.87	250.12	848.82	

10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.57	26.01	242.05	821.43	1,080,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.47	0.00	0.00	0.00	6,263,550.00 *gal*
Application event totals		55.04	26.01	242.05	821.43	

Field 4 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 4

Crop: Wheat, silage, soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 11/05/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.27	12.55	185.56	1,009.70	1,242,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.96	0.00	0.00	0.00	12,089,325.00 *gal*
Application event totals		96.23	12.55	185.56	1,009.70	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.62	13.29	207.95	1,187.82	1,236,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.94	0.00	0.00	0.00	12,055,650.00 *gal*
Application event totals		119.56	13.29	207.95	1,187.82	

Field 4 - 05/22/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	185.35	60.14	770.03	0.00	900.00 *ton*
Application event totals		185.35	60.14	770.03	0.00	

05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.28	10.40	119.90	486.73	882,000.00 *gal*
Irrigation Water - Canal Water	Surface water	4.46	0.00	0.00	0.00	9,058,575.00 *gal*
Application event totals		36.74	10.40	119.90	486.73	

Field 4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.28	0.00	0.00	0.00	8,688,150.00 *gal*
Application event totals		4.28	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.03	0.00	0.00	0.00	8,183,025.00 *gal*
Application event totals		4.03	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.50	10.47	120.72	490.04	888,000.00 *gal*
Irrigation Water - Canal Water	Surface water	3.82	0.00	0.00	0.00	7,745,250.00 *gal*
Application event totals		36.31	10.47	120.72	490.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.30	9.76	112.56	456.93	828,000.00 *gal*
Irrigation Water - Canal Water	Surface water	3.83	0.00	0.00	0.00	7,778,925.00 *gal*
Application event totals		34.13	9.76	112.56	456.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.85	0.00	0.00	0.00	7,812,600.00 *gal*
Application event totals		3.85	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.06	0.00	0.00	0.00	8,250,375.00 *gal*
Application event totals		4.06	0.00	0.00	0.00	

Field 4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.26	0.00	0.00	0.00	8,654,475.00 *gal*
Application event totals		4.26	0.00	0.00	0.00	

Field 4 - 09/22/2023: Sudangrass, silage

Field name: Field 4

Crop: Sudangrass, silage

Plant date: 09/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.76	18.76	174.60	592.52	1,380,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.76	0.00	0.00	0.00	11,685,225.00 *gal*
Application event totals		41.51	18.76	174.60	592.52	

10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	50.06	16.13	185.97	754.93	1,368,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.72	0.00	0.00	0.00	11,617,875.00 *gal*
Application event totals		55.79	16.13	185.97	754.93	

Field 5 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 5

Crop: Wheat, silage, soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 5 - 11/05/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	104.69	14.56	215.20	1,171.01	999,000.00 *gal*
Irrigation Water - Canal Water	Surface water	6.00	0.00	0.00	0.00	8,452,425.00 *gal*
Application event totals		110.70	14.56	215.20	1,171.01	

03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.75	14.43	213.26	1,160.46	990,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.91	0.00	0.00	0.00	8,317,725.00 *gal*
Application event totals		109.66	14.43	213.26	1,160.46	

Field 5 - 05/22/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	100.96	32.76	419.44	0.00	340.00 *ton*
Application event totals		100.96	32.76	419.44	0.00	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.80	0.00	0.00	0.00	5,354,325.00 *gal*
Application event totals		3.80	0.00	0.00	0.00	

Field 5 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.26	13.25	195.81	1,065.51	909,000.00 *gal*
Irrigation Water - Canal Water	Surface water	3.83	0.00	0.00	0.00	5,388,000.00 *gal*
Application event totals		99.09	13.25	195.81	1,065.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.49	15.30	176.41	716.12	900,000.00 *gal*
Irrigation Water - Canal Water	Surface water	3.73	0.00	0.00	0.00	5,253,300.00 *gal*
Application event totals		51.22	15.30	176.41	716.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.83	0.00	0.00	0.00	5,388,000.00 *gal*
Application event totals		3.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.73	0.00	0.00	0.00	5,253,300.00 *gal*
Application event totals		3.73	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.80	0.00	0.00	0.00	5,354,325.00 *gal*
Application event totals		3.80	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.95	0.00	0.00	0.00	5,556,375.00 *gal*
Application event totals		3.95	0.00	0.00	0.00	

Field 5 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.87	0.00	0.00	0.00	5,455,350.00 *gal*
Application event totals		3.87	0.00	0.00	0.00	

Field 5 - 09/22/2023: Sudangrass, silage

Field name: Field 5

Crop: Sudangrass, silage

Plant date: 09/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.34	20.64	192.09	651.89	1,053,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.67	0.00	0.00	0.00	7,980,975.00 *gal*
Application event totals		45.01	20.64	192.09	651.89	

10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.09	17.75	204.63	830.70	1,044,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.57	0.00	0.00	0.00	7,846,275.00 *gal*
Application event totals		60.66	17.75	204.63	830.70	

Field 6 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 6

Crop: Wheat, silage, soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 6 - 11/05/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	88.56	28.73	367.90	0.00	430.00 *ton*
Application event totals		88.56	28.73	367.90	0.00	

02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.26	13.94	160.68	652.28	1,182,000.00 *gal*
Irrigation Water - Canal Water	Surface water	6.07	0.00	0.00	0.00	12,325,050.00 *gal*
Application event totals		49.33	13.94	160.68	652.28	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.17	11.71	173.01	941.41	1,158,000.00 *gal*
Irrigation Water - Canal Water	Surface water	6.06	0.00	0.00	0.00	12,291,375.00 *gal*
Application event totals		90.22	11.71	173.01	941.41	

Field 6 - 05/22/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	150.34	48.78	624.58	0.00	730.00 *ton*
Application event totals		150.34	48.78	624.58	0.00	

Field 6 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.81	11.10	164.04	892.64	1,098,000.00 *gal*
Irrigation Water - Canal Water	Surface water	4.10	0.00	0.00	0.00	8,317,725.00 *gal*
Application event totals		83.90	11.10	164.04	892.64	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.05	0.00	0.00	0.00	8,216,700.00 *gal*
Application event totals		4.05	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.06	0.00	0.00	0.00	8,250,375.00 *gal*
Application event totals		4.06	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.84	13.16	151.71	615.86	1,116,000.00 *gal*
Irrigation Water - Canal Water	Surface water	3.93	0.00	0.00	0.00	7,980,975.00 *gal*
Application event totals		44.77	13.16	151.71	615.86	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.82	0.00	0.00	0.00	7,745,250.00 *gal*
Application event totals		3.82	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.83	0.00	0.00	0.00	7,778,925.00 *gal*
Application event totals		3.83	0.00	0.00	0.00	

Field 6 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.85	0.00	0.00	0.00	7,812,600.00 *gal*
Application event totals		3.85	0.00	0.00	0.00	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	3.78	0.00	0.00	0.00	7,677,900.00 *gal*
Application event totals		3.78	0.00	0.00	0.00	

Field 6 - 09/22/2023: Sudangrass, silage

Field name: Field 6

Crop: Sudangrass, silage

Plant date: 09/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.84	19.33	179.91	610.56	1,422,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.62	0.00	0.00	0.00	11,415,825.00 *gal*
Application event totals		42.47	19.33	179.91	610.56	

10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.48	16.91	194.94	791.35	1,434,000.00 *gal*
Irrigation Water - Canal Water	Surface water	5.59	0.00	0.00	0.00	11,348,475.00 *gal*
Application event totals		58.07	16.91	194.94	791.35	

Field 7 - 11/05/2022: Wheat, silage, soft dough

Field 7 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 7

Crop: Wheat, silage, soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	198.62	64.44	825.17	0.00	280.00 *ton*
Application event totals		198.62	64.44	825.17	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	6.00	0.00	0.00	0.00	3,535,875.00 *gal*
Application event totals		6.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	5.89	0.00	0.00	0.00	3,468,525.00 *gal*
Application event totals		5.89	0.00	0.00	0.00	

Field 7 - 05/22/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	191.53	62.14	795.70	0.00	270.00 *ton*
Application event totals		191.53	62.14	795.70	0.00	

Field 7 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.11	0.00	0.00	0.00	2,424,600.00 *gal*
Application event totals		4.11	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.34	0.00	0.00	0.00	2,559,300.00 *gal*
Application event totals		4.34	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.29	0.00	0.00	0.00	2,525,625.00 *gal*
Application event totals		4.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Irrigation Water - Canal Water	Surface water	4.11	0.00	0.00	0.00	2,424,600.00 *gal*
Application event totals		54.11	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.11	0.00	0.00	0.00	2,424,600.00 *gal*
Application event totals		4.11	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.29	0.00	0.00	0.00	2,525,625.00 *gal*
Application event totals		4.29	0.00	0.00	0.00	

Field 7 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	50.00	0.00	0.00	0.00	
Irrigation Water - Canal Water	Surface water	4.34	0.00	0.00	0.00	2,559,300.00 *gal*
Application event totals		54.34	0.00	0.00	0.00	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	4.23	0.00	0.00	0.00	2,491,950.00 *gal*
Application event totals		4.23	0.00	0.00	0.00	

Field 7 - 09/22/2023: Sudangrass, silage

Field name: Field 7

Crop: Sudangrass, silage

Plant date: 09/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	113.50	36.83	471.52	0.00	160.00 *ton*
Application event totals		113.50	36.83	471.52	0.00	

09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	5.89	0.00	0.00	0.00	3,468,525.00 *gal*
Application event totals		5.89	0.00	0.00	0.00	

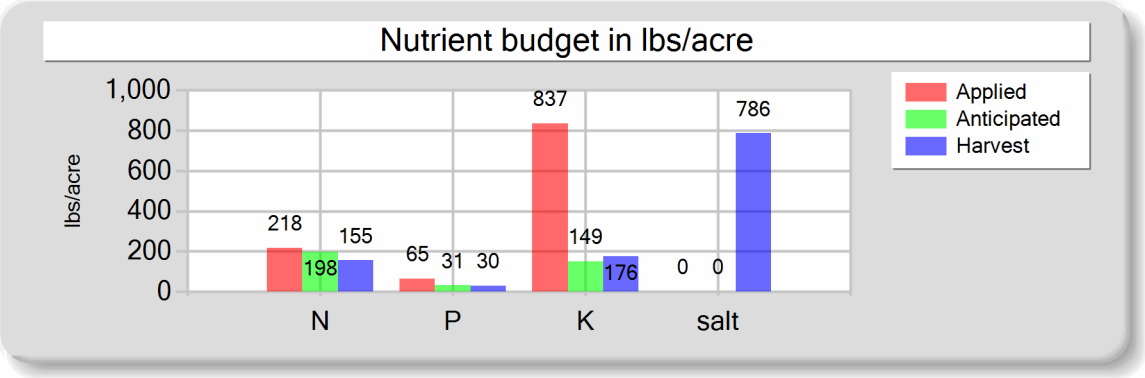
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal Water	Surface water	5.83	0.00	0.00	0.00	3,434,850.00 *gal*
Application event totals		5.83	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field 1 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 1 Crop: Wheat, silage, soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	201.35	65.33	836.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	12.03	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	218.04	65.33	836.50	0.00
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	155.31	29.59	176.30	786.00
Nutrient balance	62.73	35.74	660.20	-786.00
Applied to removed ratio	1.40	2.21	4.74	0.00

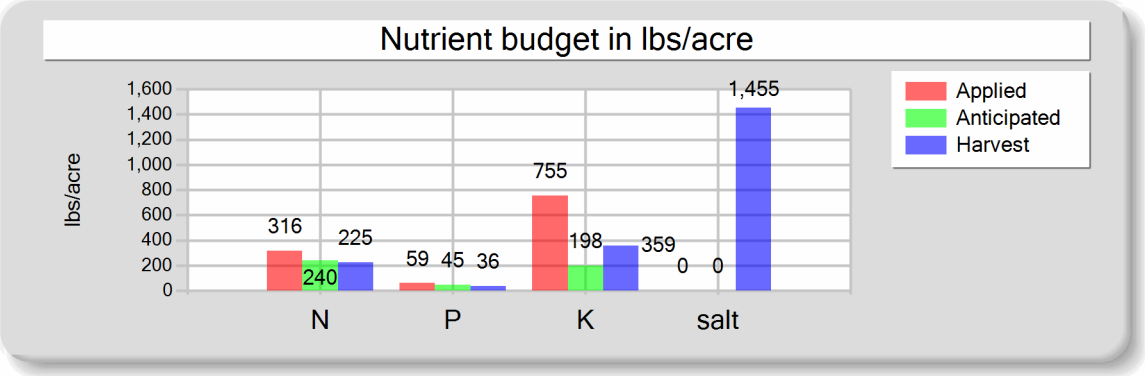
Fresh water applied	
10,237,200.00	*gallons*
377.00	*acre-inches*
14.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/22/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	100.00	0.00	0.00	0.00
Dry manure	181.70	58.96	754.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	29.31	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	315.68	58.96	754.89	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.41	35.95	359.49	1,455.45
Nutrient balance	90.27	23.01	395.40	-1,455.45
Applied to removed ratio	1.40	1.64	2.10	0.00

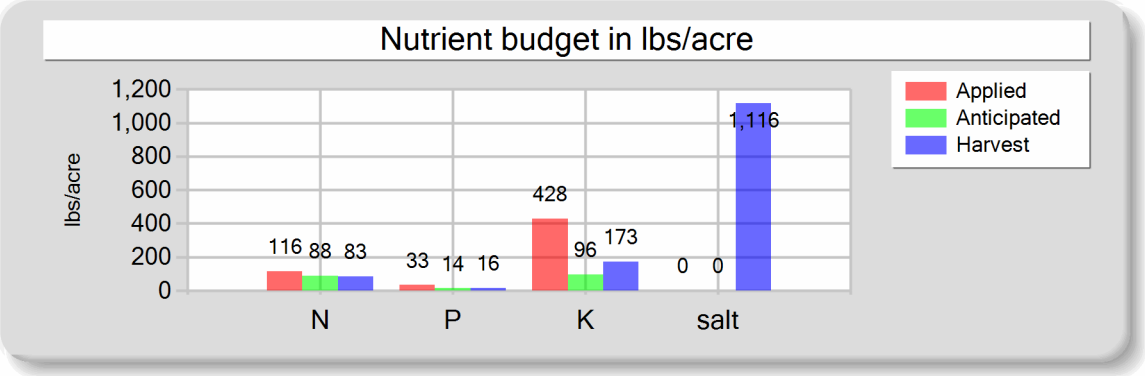
Fresh water applied	
24,953,175.00	*gallons*
918.94	*acre-inches*
35.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 09/22/2023: Sudangrass, silage

Field name: Field 1 Crop: Sudangrass, silage Plant date: 09/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.13	33.46	428.45	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.31	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	116.10	33.46	428.45	0.00
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	83.32	16.39	173.17	1,116.34
Nutrient balance	32.79	17.07	255.28	-1,116.34
Applied to removed ratio	1.39	2.04	2.47	0.00

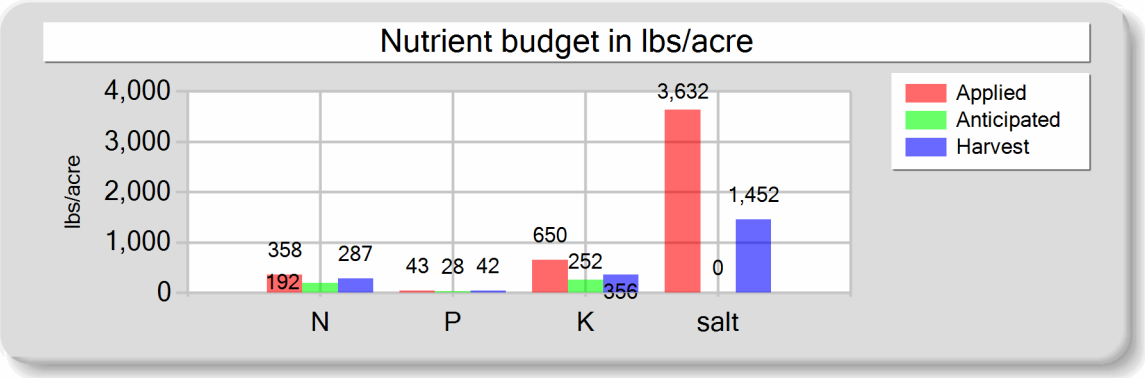
Fresh water applied	
7,071,750.00	*gallons*
260.43	*acre-inches*
10.02	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/05/2022: Ryegrass, hay

Field name: Field 2 Crop: Ryegrass, hay Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.12	42.70	650.23	3,632.08
Fresh water	13.89	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	358.01	42.70	650.23	3,632.08
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	287.09	42.25	356.41	1,451.75
Nutrient balance	70.92	0.45	293.81	2,180.32
Applied to removed ratio	1.25	1.01	1.82	2.50

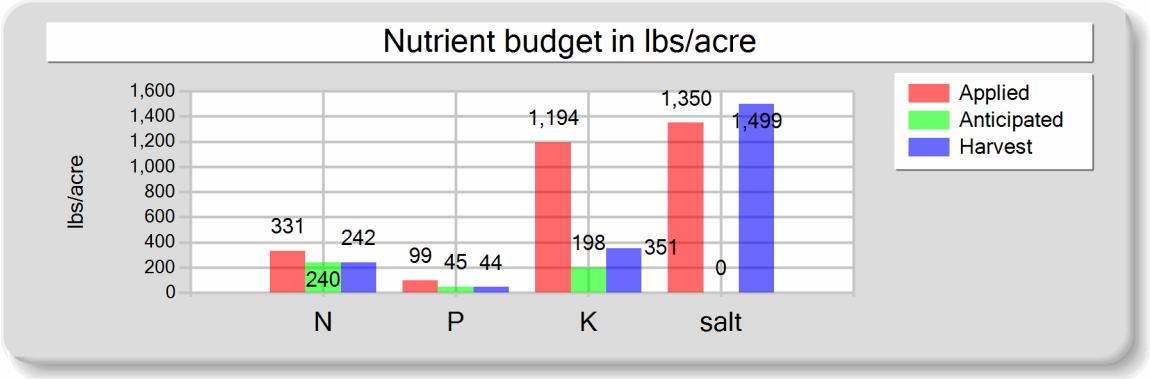
Fresh water applied	
10,001,475.00	*gallons*
368.32	*acre-inches*
16.74	*inches/acre*

Process wastewater applied	
1,452,000.00	*gallons*
53.47	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/22/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.13	65.91	843.92	0.00
Process wastewater	87.32	32.66	350.47	1,350.05
Fresh water	33.19	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	330.65	98.57	1,194.39	1,350.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	242.16	43.67	351.33	1,498.60
Nutrient balance	88.49	54.90	843.06	-148.55
Applied to removed ratio	1.37	2.26	3.40	0.90

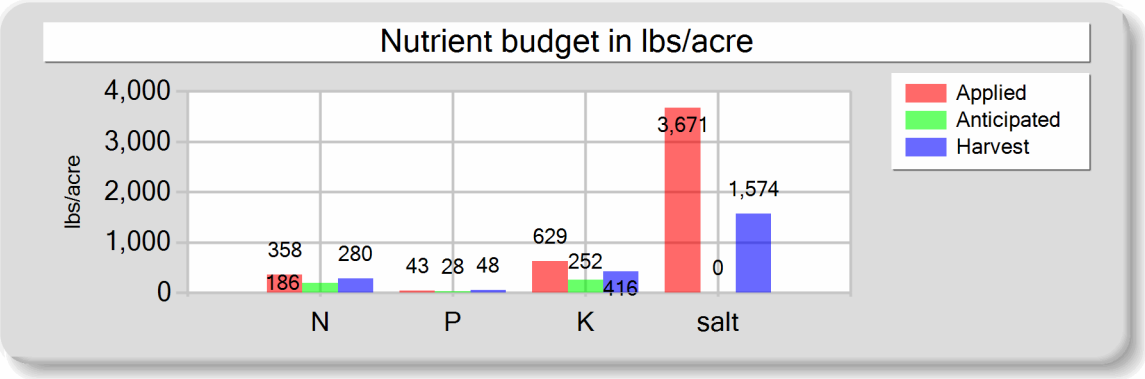
Fresh water applied	
23,909,250.00	*gallons*
880.50	*acre-inches*
40.02	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/05/2022: Ryegrass, hay

Field name: Field 3 Crop: Ryegrass, hay Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	340.10	43.28	628.81	3,671.19
Fresh water	13.28	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	358.05	43.28	628.81	3,671.19
Anticipated crop nutrient removal	186.00	27.60	252.00	0.00
Actual crop nutrient removal	279.78	48.04	415.62	1,574.33
Nutrient balance	78.27	-4.75	213.19	2,096.86
Applied to removed ratio	1.28	0.90	1.51	2.33

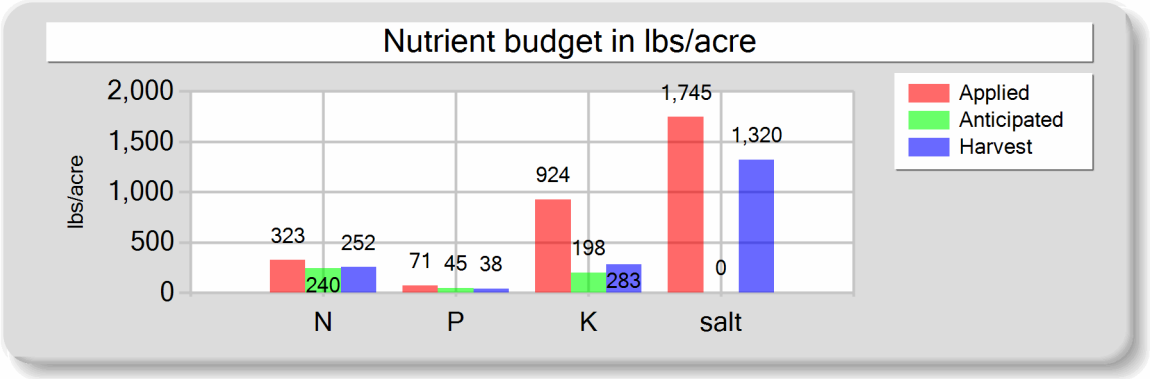
Fresh water applied	
15,221,100.00	*gallons*
560.54	*acre-inches*
16.02	*inches/acre*

Process wastewater applied	
2,346,000.00	*gallons*
86.40	*acre-inches*
2.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/22/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	138.63	44.98	575.93	0.00
Process wastewater	145.82	25.62	348.14	1,744.64
Fresh water	34.32	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	323.44	70.60	924.08	1,744.64
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.21	37.63	282.72	1,320.05
Nutrient balance	71.23	32.97	641.35	424.59
Applied to removed ratio	1.28	1.88	3.27	1.32

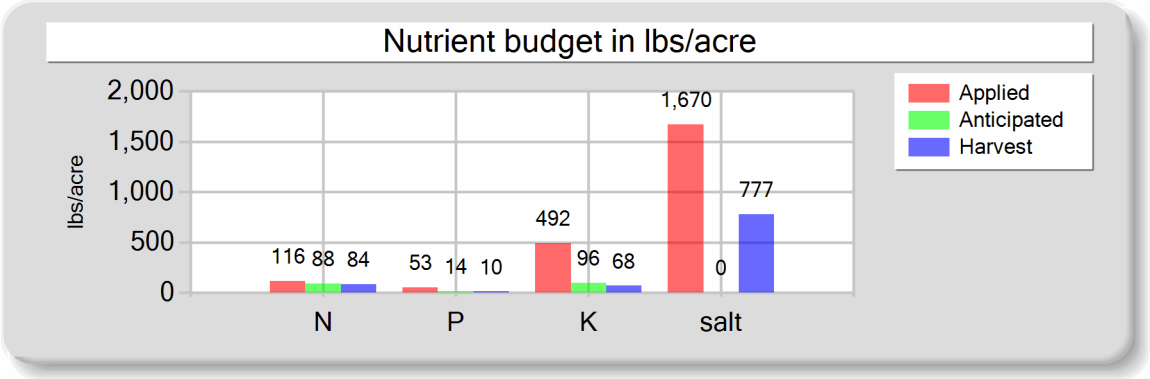
Fresh water applied	
39,332,400.00	*gallons*
1,448.48	*acre-inches*
41.39	*inches/acre*

Process wastewater applied	
1,356,000.00	*gallons*
49.94	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 09/22/2023: Sudangrass, silage

Field name: Field 3 Crop: Sudangrass, silage Plant date: 09/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	100.79	52.88	492.17	1,670.25
Fresh water	11.02	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	116.48	52.88	492.17	1,670.25
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	83.98	10.45	67.98	776.94
Nutrient balance	32.50	42.43	424.19	893.31
Applied to removed ratio	1.39	5.06	7.24	2.15

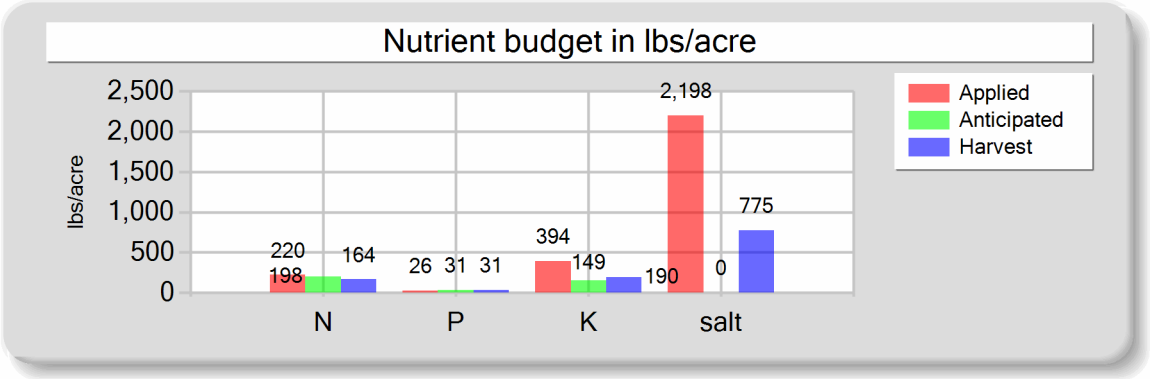
Fresh water applied	
12,628,125.00	*gallons*
465.05	*acre-inches*
13.29	*inches/acre*

Process wastewater applied	
2,196,000.00	*gallons*
80.87	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 4 Crop: Wheat, silage, soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.90	25.85	393.51	2,197.52
Fresh water	11.89	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	220.46	25.85	393.51	2,197.52
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	163.83	31.11	189.75	774.57
Nutrient balance	56.63	-5.26	203.76	1,422.96
Applied to removed ratio	1.35	0.83	2.07	2.84

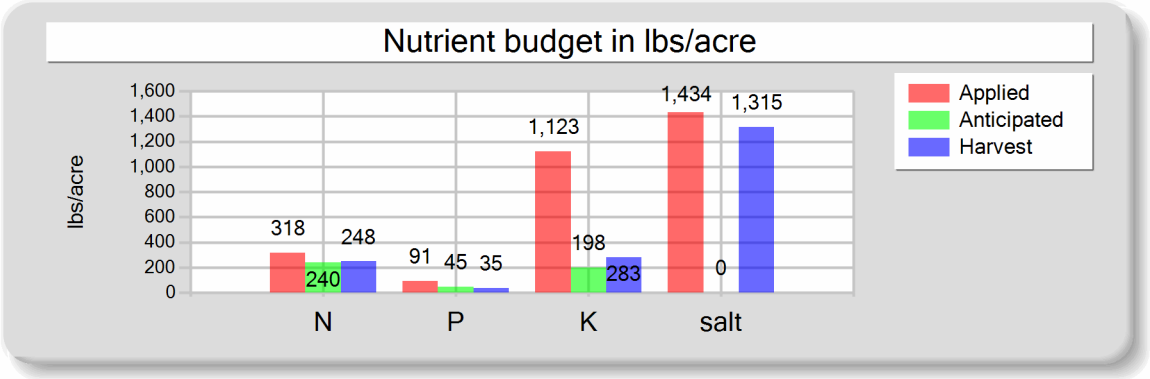
Fresh water applied	
24,144,975.00	*gallons*
889.18	*acre-inches*
14.34	*inches/acre*

Process wastewater applied	
2,478,000.00	*gallons*
91.26	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/22/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	185.35	60.14	770.03	0.00
Process wastewater	95.08	30.63	353.18	1,433.70
Fresh water	32.60	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	317.69	90.77	1,123.21	1,433.70
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.71	35.39	283.10	1,315.25
Nutrient balance	69.98	55.38	840.11	118.45
Applied to removed ratio	1.28	2.57	3.97	1.09

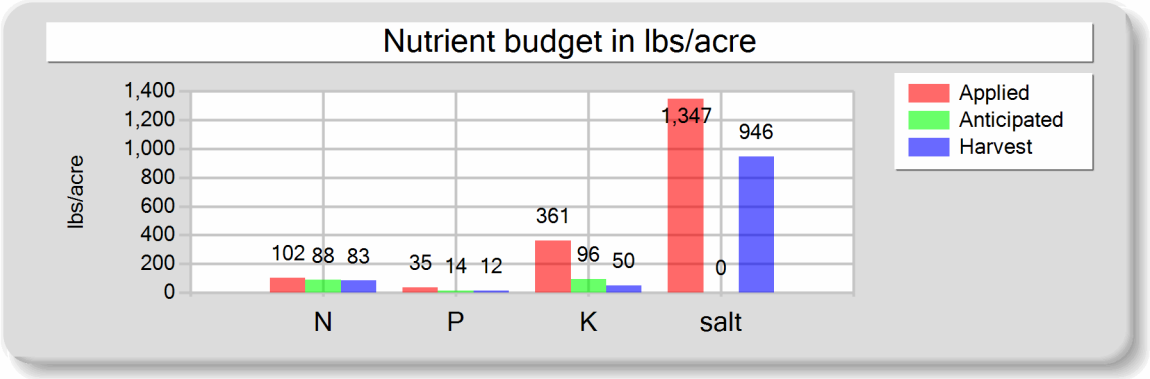
Fresh water applied	
66,171,375.00	*gallons*
2,436.87	*acre-inches*
39.30	*inches/acre*

Process wastewater applied	
2,598,000.00	*gallons*
95.68	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 09/22/2023: Sudangrass, silage

Field name: Field 4 Crop: Sudangrass, silage Plant date: 09/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	85.82	34.89	360.57	1,347.45
Fresh water	11.48	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	101.97	34.89	360.57	1,347.45
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	83.44	12.44	50.22	945.68
Nutrient balance	18.52	22.45	310.35	401.77
Applied to removed ratio	1.22	2.80	7.18	1.42

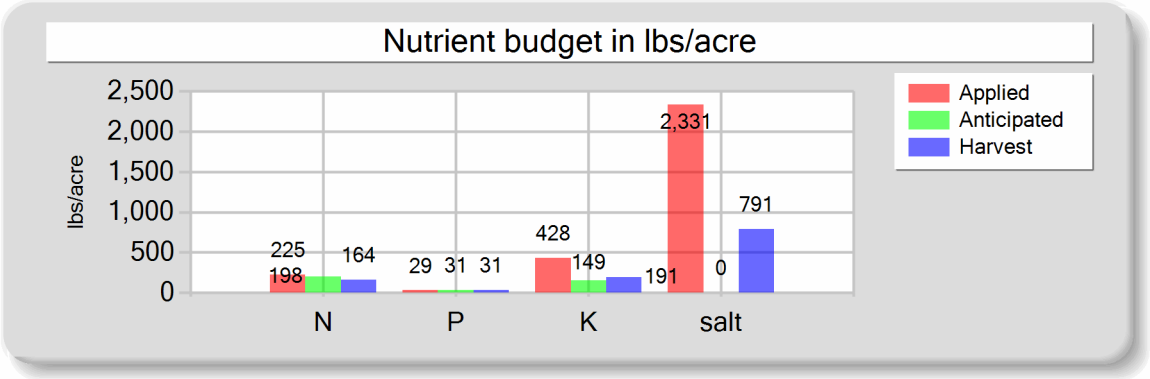
Fresh water applied	
23,303,100.00	*gallons*
858.17	*acre-inches*
13.84	*inches/acre*

Process wastewater applied	
2,748,000.00	*gallons*
101.20	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 5 Crop: Wheat, silage, soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	208.44	28.99	428.47	2,331.47
Fresh water	11.91	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	225.02	28.99	428.47	2,331.47
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	163.68	30.63	190.79	791.28
Nutrient balance	61.34	-1.64	237.67	1,540.19
Applied to removed ratio	1.37	0.95	2.25	2.95

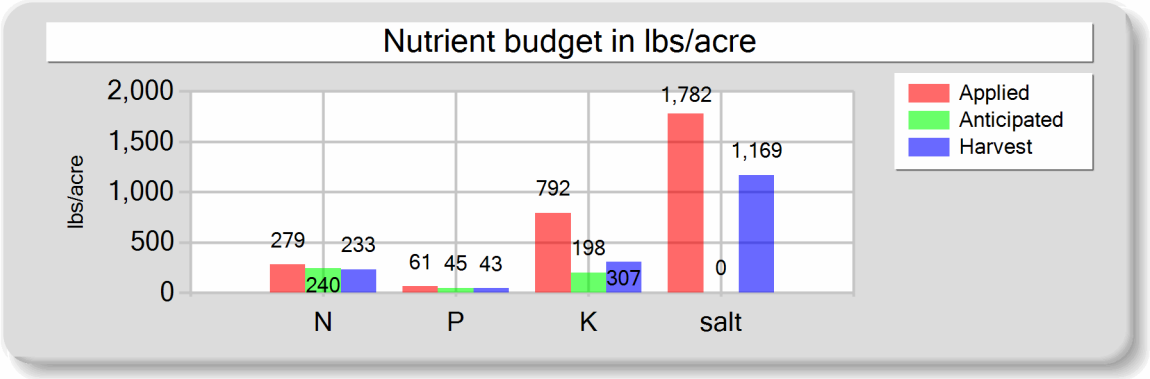
Fresh water applied	
16,770,150.00	*gallons*
617.59	*acre-inches*
14.36	*inches/acre*

Process wastewater applied	
1,989,000.00	*gallons*
73.25	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/22/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	100.96	32.76	419.44	0.00
Process wastewater	142.75	28.55	372.22	1,781.63
Fresh water	30.54	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	278.92	61.31	791.66	1,781.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	232.76	43.01	307.25	1,169.40
Nutrient balance	46.16	18.29	484.41	612.23
Applied to removed ratio	1.20	1.43	2.58	1.52

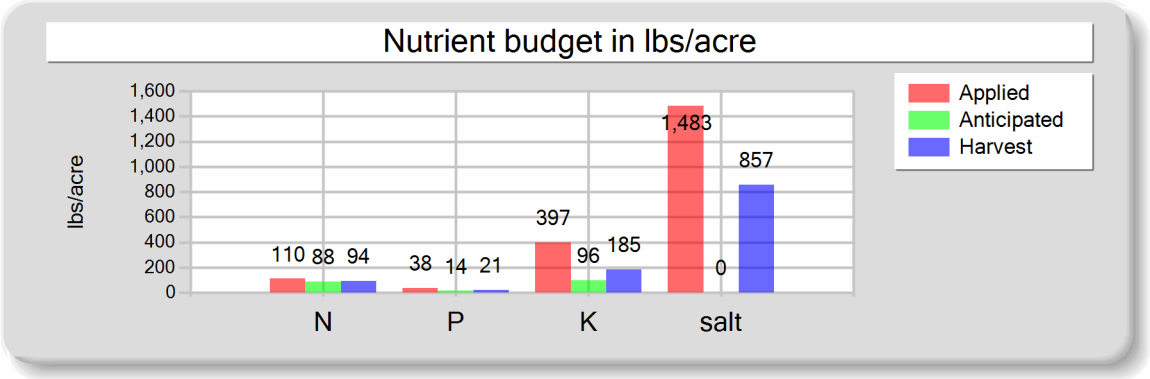
Fresh water applied	
43,002,975.00	*gallons*
1,583.65	*acre-inches*
36.83	*inches/acre*

Process wastewater applied	
1,809,000.00	*gallons*
66.62	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 09/22/2023: Sudangrass, silage

Field name: Field 5 Crop: Sudangrass, silage Plant date: 09/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.43	38.39	396.73	1,482.59
Fresh water	11.24	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	110.34	38.39	396.73	1,482.59
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	93.86	20.92	184.82	857.47
Nutrient balance	16.48	17.47	211.91	625.12
Applied to removed ratio	1.18	1.84	2.15	1.73

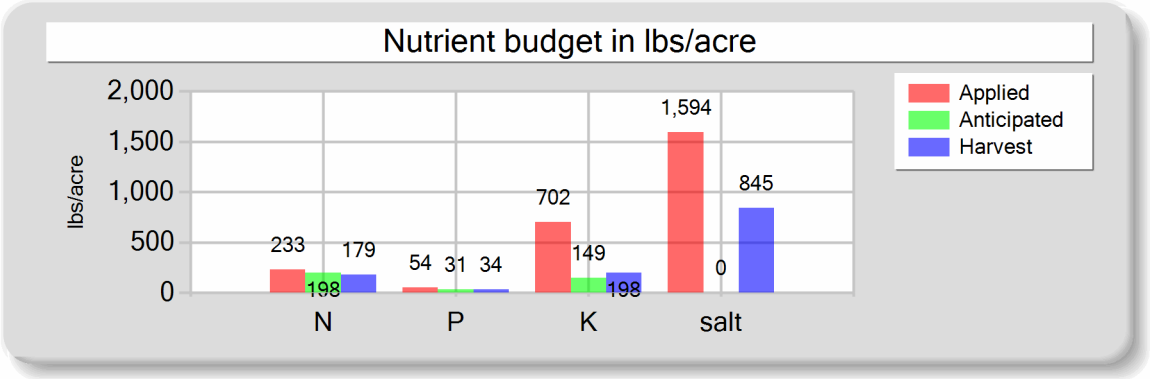
Fresh water applied	
15,827,250.00	*gallons*
582.86	*acre-inches*
13.55	*inches/acre*

Process wastewater applied	
2,097,000.00	*gallons*
77.23	*acre-inches*
1.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 6 Crop: Wheat, silage, soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	88.56	28.73	367.90	0.00
Process wastewater	127.42	25.64	333.69	1,593.70
Fresh water	12.13	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	232.77	54.37	701.59	1,593.70
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	179.06	33.93	197.91	844.67
Nutrient balance	53.71	20.45	503.68	749.03
Applied to removed ratio	1.30	1.60	3.54	1.89

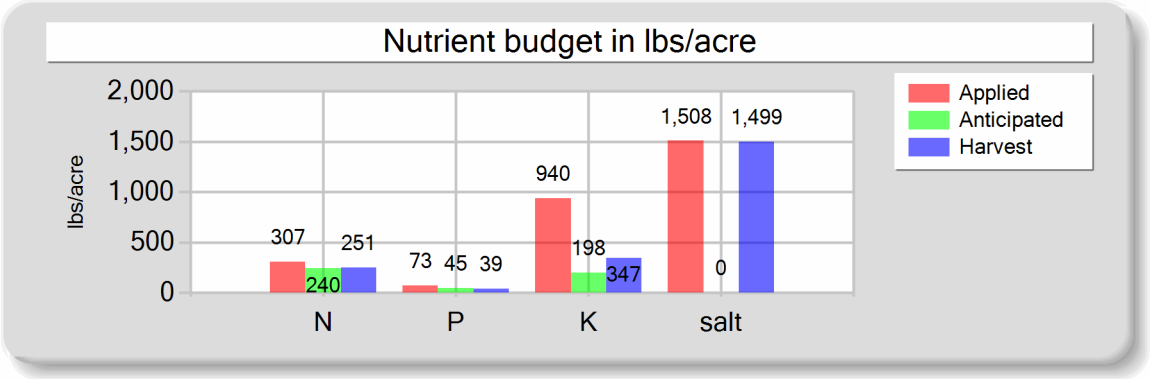
Fresh water applied	
24,616,425.00	*gallons*
906.54	*acre-inches*
14.62	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 05/22/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	150.34	48.78	624.58	0.00
Process wastewater	120.65	24.26	315.76	1,508.50
Fresh water	31.42	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	307.07	73.04	940.33	1,508.50
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	250.50	38.66	347.16	1,499.08
Nutrient balance	56.57	34.38	593.18	9.42
Applied to removed ratio	1.23	1.89	2.71	1.01

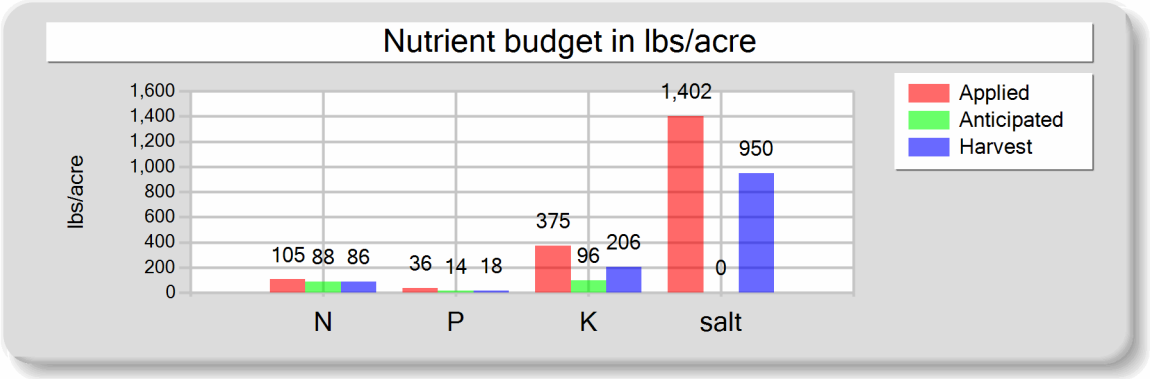
Fresh water applied	
63,780,450.00	*gallons*
2,348.82	*acre-inches*
37.88	*inches/acre*

Process wastewater applied	
2,214,000.00	*gallons*
81.53	*acre-inches*
1.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 09/22/2023: Sudangrass, silage

Field name: Field 6 Crop: Sudangrass, silage Plant date: 09/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	89.32	36.24	374.85	1,401.90
Fresh water	11.21	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	105.20	36.24	374.85	1,401.90
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	85.88	17.79	206.48	950.16
Nutrient balance	19.32	18.45	168.38	451.74
Applied to removed ratio	1.23	2.04	1.82	1.48

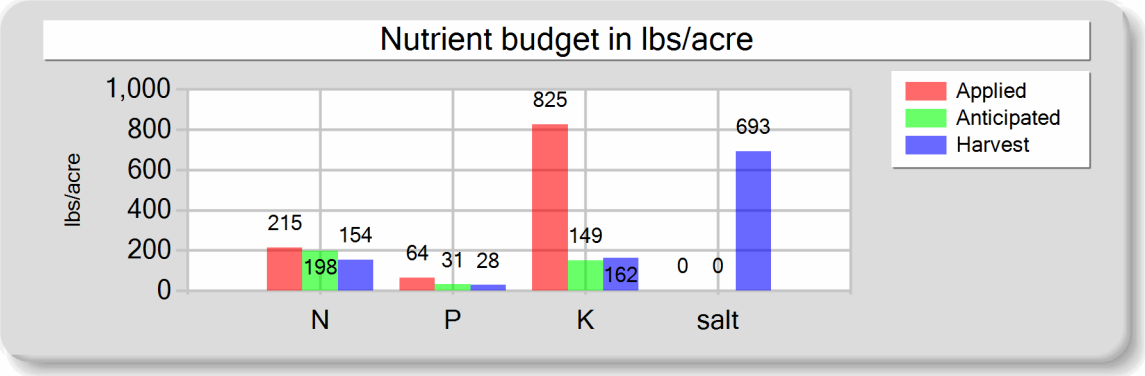
Fresh water applied	
22,764,300.00	*gallons*
838.33	*acre-inches*
13.52	*inches/acre*

Process wastewater applied	
2,856,000.00	*gallons*
105.18	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/05/2022: Wheat, silage, soft dough

Field name: Field 7 Crop: Wheat, silage, soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	198.62	64.44	825.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.89	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	215.17	64.44	825.17	0.00
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	153.69	27.60	161.78	692.62
Nutrient balance	61.48	36.84	663.39	-692.62
Applied to removed ratio	1.40	2.33	5.10	0.00

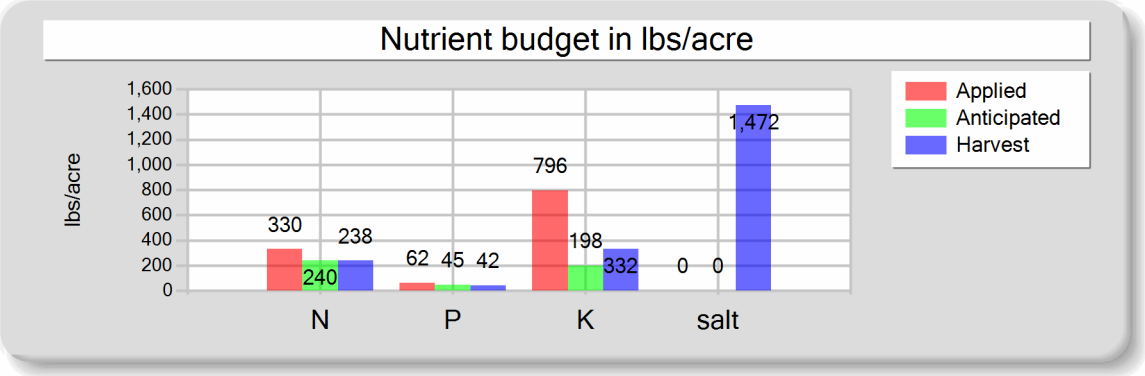
Fresh water applied	
7,004,400.00	*gallons*
257.95	*acre-inches*
14.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 05/22/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	100.00	0.00	0.00	0.00
Dry manure	191.53	62.14	795.70	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	33.83	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	330.02	62.14	795.70	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	237.92	42.34	331.59	1,472.46
Nutrient balance	92.10	19.81	464.11	-1,472.46
Applied to removed ratio	1.39	1.47	2.40	0.00

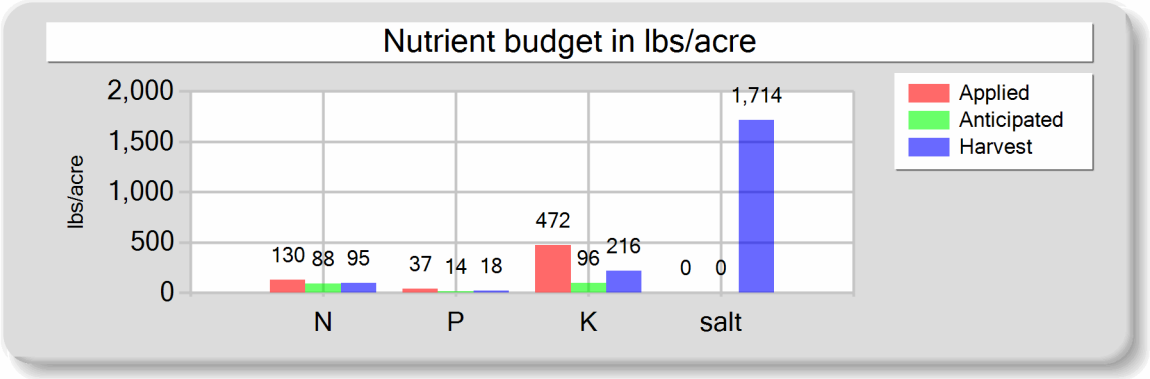
Fresh water applied	
19,935,600.00	*gallons*
734.16	*acre-inches*
40.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 09/22/2023: Sudangrass, silage

Field name: Field 7 Crop: Sudangrass, silage Plant date: 09/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	113.50	36.83	471.52	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.71	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	129.88	36.83	471.52	0.00
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	94.96	17.80	216.29	1,714.49
Nutrient balance	34.92	19.02	255.23	-1,714.49
Applied to removed ratio	1.37	2.07	2.18	0.00

Fresh water applied	
6,903,375.00	*gallons*
254.23	*acre-inches*
14.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 5.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,900.00	4,100.00	1,500.00							
DL	10.00	10.00	10.00							

Solid Manure

Sample and source description: Solid Manure
Sample date: 09/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,300.00	7,560.00	96,800.00							
DL	0.10	0.10	0.10							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	271.00	110.00	0.00	0.90	87.60	1,010.00								5,410.00	4,100
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	192.00	88.90	0.00	0.50	101.00	940.00								5,220.00	3,190
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	540.00	325.00	0.00	0.00	75.10	1,110.00								8,080.00	6,040
DL	0.10	0.10	0.10	0.10	0.10	0.10								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	683.00	400.00	0.00	0.00	79.90	1,250.00								9,350.00	7,140
DL	0.10	0.10	0.10	0.10	0.10	0.10								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW

Domestic Well - DW

DW

Sample description: DW

Sample date: 11/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	22.20	0.00	22.20								1,250.00	0
DL	0.10	0.10	0.10								10.00	1

Irrigation Water - Canal Water

Canal Water

Sample description: Canal Water

Sample date: 09/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.66	0.00	3.66								0.00	0
DL	0.10	0.10	0.10								0.10	1

D. SOIL ANALYSES

Field 1

Field 1

Sample and source description: Field 1

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			50.80				
DL			0.10				

Field 2

Field 2

Sample and source description: Field 2

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			53.10				
DL			0.10				

Field 3

Field 3

Sample and source description: Field 3

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.90				
DL			0.10				

Field 4

Field 4

Sample and source description: Field 4

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			69.10				
DL			0.10				

Field 5

Field 5

Field 5

Sample and source description: Field 5

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			65.30				
DL			0.10				

Field 6

Field 6

Sample and source description: Field 6

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			48.20				
DL			0.10				

Field 7

Field 7

Sample and source description: Field 7

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			51.10				
DL			0.10				

E. PLANT TISSUE ANALYSES

Field 1 - 11/05/2022: Wheat, silage, soft dough

Field 1 - 11/05/2022: Wheat, silage, soft dough

Field 1

Sample and source description: Field 1

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	2,820.00	16,800.00		7.49
DL	10.00	10.00	10.00		1.00

Field 1 - 05/22/2023: Corn, silage

Field 1

Sample and source description: Field 1

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	1,850.00	18,500.00		7.49
DL	10.00	10.00	10.00		1.00

Field 1 - 09/22/2023: Sudangrass, silage

Field 1A

Sample and source description: Field 1A

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	3,010.00	31,800.00		20.50
DL	10.00	10.00	10.00		10.00

Field 1 - 09/22/2023: Sudangrass, silage

Field 1B

Sample and source description: Field 1B

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	2,900.00	34,700.00		23.40
DL	10.00	10.00	10.00		10.00

Field 2 - 11/05/2022: Ryegrass, hay

Field 2

Sample and source description: Field 2

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,200.00	5,180.00	43,700.00		17.80
DL	10.00	10.00	10.00		10.00

Field 2 - 05/22/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	2,200.00	17,700.00		7.55
DL	10.00	10.00	10.00		1.00

Field 3 - 11/05/2022: Ryegrass, hay

Field 3

Sample and source description: Field 3

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,100.00	5,340.00	46,200.00		17.50
DL	10.00	10.00	10.00		10.00

Field 3 - 05/22/2023: Corn, silage

Field 3

Sample and source description: Field 3

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	1,850.00	13,900.00		6.49
DL	10.00	10.00	10.00		1.00

Field 3 - 09/22/2023: Sudangrass, silage

Field 3C

Sample and source description: Field 3C

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	1,830.00	11,900.00		13.60
DL	10.00	10.00	10.00		10.00

Field 4 - 11/05/2022: Wheat, silage, soft dough

Field 4A

Sample and source description: Field 4A

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	3,000.00	18,300.00		7.47
DL	10.00	10.00	10.00		1.00

Field 4B

Sample and source description: Field 4B

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,000.00	2,700.00	15,800.00		7.48
DL	10.00	10.00	10.00		1.00

Field 4 - 05/22/2023: Corn, silage

Field 4A

Sample and source description: Field 4A

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	1,800.00	14,400.00		6.69
DL	10.00	10.00	10.00		1.00

Field 4 - 05/22/2023: Corn, silage

Field 4B

Sample and source description: Field 4B

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,160.00	16,700.00		7.87
DL	10.00	10.00	10.00		1.00

Field 4 - 09/22/2023: Sudangrass, silage

Field 4A

Sample and source description: Field 4A

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	3,090.00	38,100.00		19.40
DL	10.00	10.00	10.00		10.00

Field 4B

Sample and source description: Field 4B

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	2,460.00	9,930.00		18.70
DL	10.00	10.00	10.00		10.00

Field 5 - 11/05/2022: Wheat, silage, soft dough

Field 5 - 11/05/2022: Wheat, silage, soft dough

Field 5

Sample and source description: Field 5

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,050.00	19,000.00		7.88
DL	10.00	10.00	10.00		1.00

Field 5 - 05/22/2023: Corn, silage

Field 5

Sample and source description: Field 5

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,310.00	16,500.00		6.28
DL	10.00	10.00	10.00		1.00

Field 5 - 09/22/2023: Sudangrass, silage

Field 5

Sample and source description: Field 5

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,610.00	31,900.00		14.80
DL	10.00	10.00	10.00		10.00

Field 6 - 11/05/2022: Wheat, silage, soft dough

Field 6A

Sample and source description: Field 6A

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,880.00	16,800.00		7.17
DL	10.00	10.00	10.00		1.00

Field 6B

Sample and source description: Field 6B

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	2,440.00	13,900.00		7.19
DL	10.00	10.00	10.00		1.00

Field 6 - 05/22/2023: Corn, silage

Field 6A

Sample and source description: Field 6A

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,090.00	15,500.00		6.34
DL	10.00	10.00	10.00		1.00

Field 6 - 05/22/2023: Corn, silage

Field 6B

Sample and source description: Field 6B

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	1,960.00	17,600.00		7.60
DL	10.00	10.00	10.00		1.00

Field 6 - 09/22/2023: Sudangrass, silage

Field 6

Sample and source description: Field 6

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	2,920.00	33,900.00		15.60
DL	10.00	10.00	10.00		10.00

Field 7 - 11/05/2022: Wheat, silage, soft dough

Field 7

Sample and source description: Field 7

Sample date: 04/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	2,730.00	16,000.00		6.85
DL	10.00	10.00	10.00		1.00

Field 7 - 05/22/2023: Corn, silage

Field 7

Sample and source description: Field 7

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,260.00	17,700.00		7.86
DL	10.00	10.00	10.00		1.00

Field 7 - 09/22/2023: Sudangrass, silage

Field 7

Sample and source description: Field 7

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,700.00	32,800.00		26.00
DL	10.00	10.00	10.00		10.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

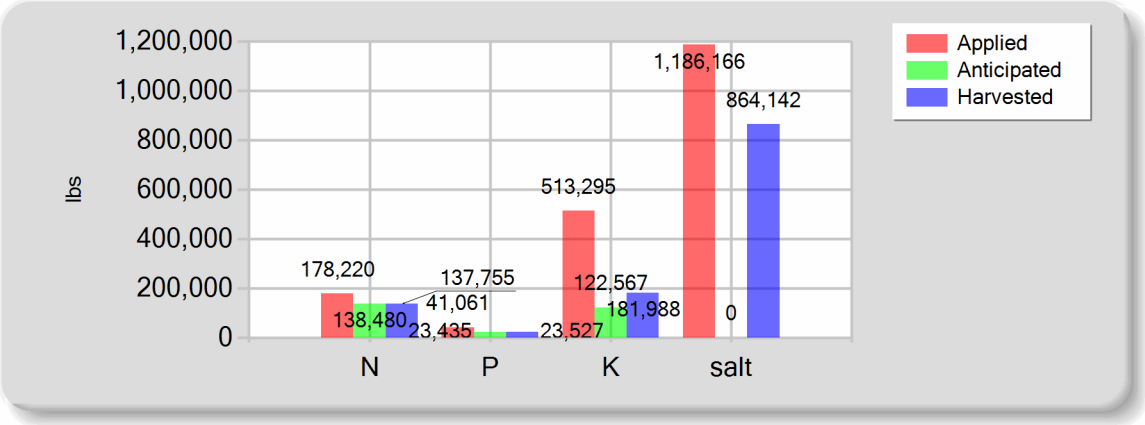
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

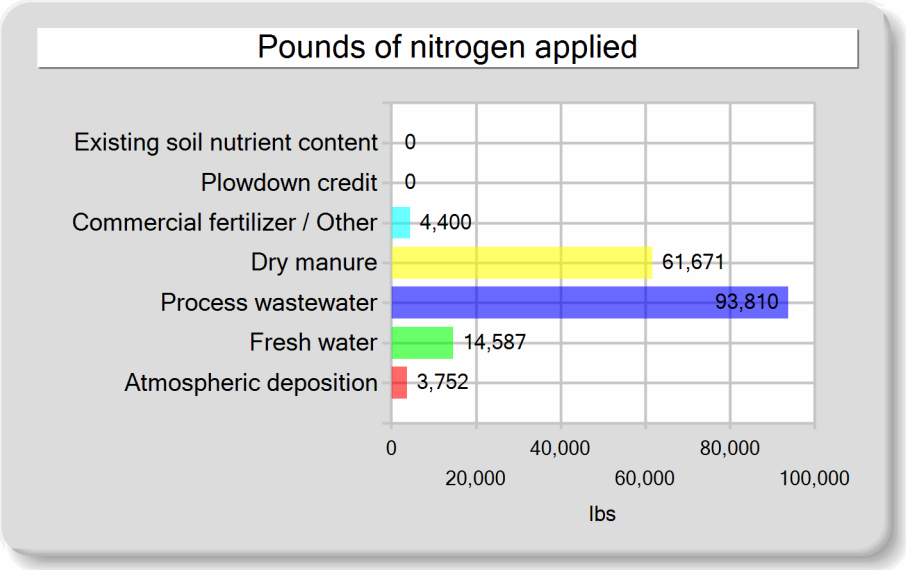
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

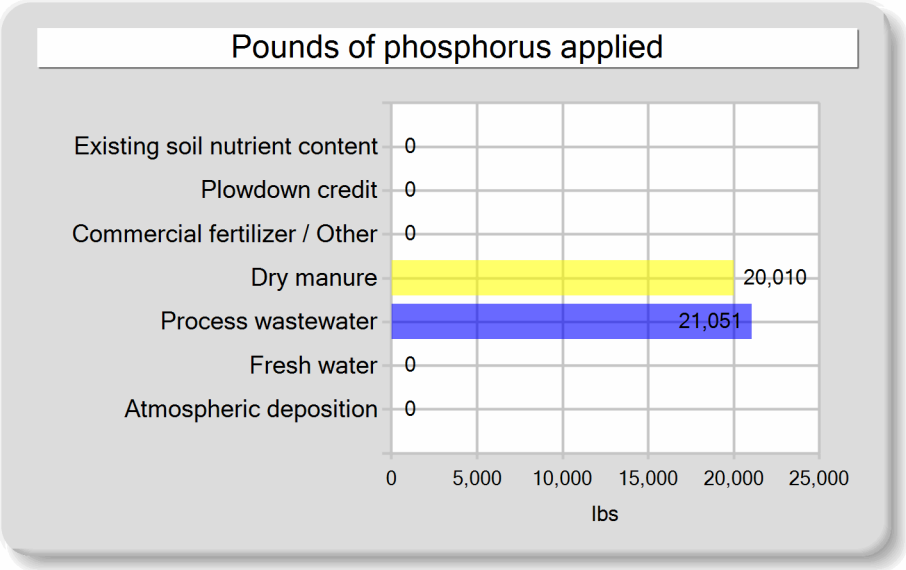
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4,400.00	0.00	0.00	0.00
Dry manure	61,671.37	20,010.11	256,214.11	0.00
Process wastewater	93,810.20	21,050.60	257,080.41	1,186,165.56
Fresh water	14,586.55	0.00	0.00	0.00
Atmospheric deposition	3,752.00	0.00	0.00	0.00
Total nutrients applied	178,220.12	41,060.72	513,294.52	1,186,165.56
Anticipated crop nutrient removal	138,480.00	23,435.40	122,567.40	0.00
Actual crop nutrient removal	137,755.49	23,527.22	181,987.67	864,141.85
Nutrient balance	40,464.63	17,533.50	331,306.85	322,023.71
Applied to removed ratio	1.29	1.75	2.82	1.37

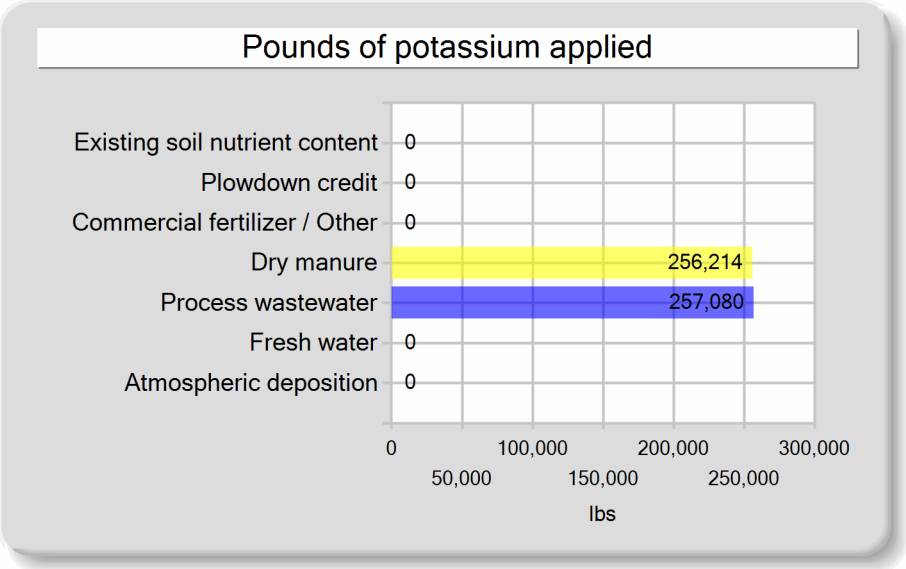
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

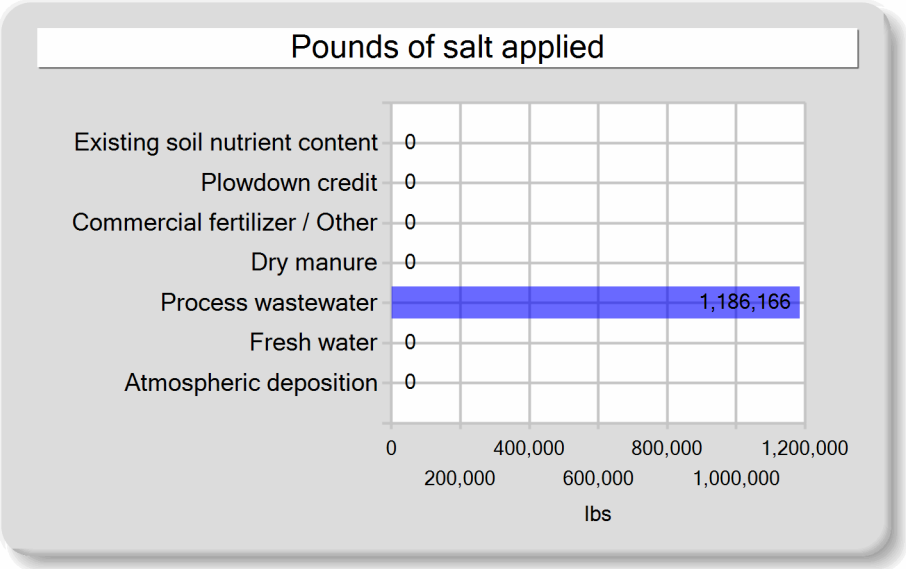


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Rye Grass Hay was harvested as SILAGE in 2023

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Rick Nutcher
Rick Nutcher (Jun 20, 2024 15:33 MDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Rick Nutcher	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/20/2024	
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Nutcher Farms
P.O. Box 268
Denair, CA 95316

Grower

Consultant:
R & J Avila
8119 Ellington Ave.
Hilmar, CA 95324

PURCHASE ORDER:
RECEIVED DATE: 11/22/23 9:51 AM
SUBMITTED BY:
ANALYZED DATE: 11/27/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Well Water	T3326004A	Electrical Conductivity (EC)	EPA 120.1 and	1.25	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	22.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	799	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6	99.4 (T3326004A)	97.6 (T3326004A)	1.82
Electrical Conductivity (EC)	95.0			0.240
Nitrate Nitrogen (NO3-N)	96.0	101 (T3326007A)	96.7 (T3326007A)	3.97

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Nutcher Farms (Rick)
Mailing address: 5312 W. GRAYSON RD.
(not facility address) MODESTO, CA. 95358
Phone#: (209) 537-1118

Authorized Copy Released To: R+J AVILA

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

Required Field Measurement

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/22/23	9Am	WELL WATER		17.6'		✓				
2	11/22/23	9Am	LAGOON WATER		14.6'			✓			
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]	Nutcher Farms		11/22/23 9:51		
2nd	[signature]	DENELE ANALYTICAL, INC.	11-22-23 9:51			T33260004

Required to be submitted to any agency? (YES / NO)